Trapped Ions Part II.

- Phonon assisted gate
- Multiple Ions
- 2-ion CNOT
- Outlook + Challenges

Phonon assisted gate

Have two coupled quantum systems (electronic + vibrational)

⟶ use this to form a gate.

Blue sidebands

$$H_{BSB} = \eta \frac{\hbar\nu}{2} \underbrace{\left[i a^\dagger \sigma_+ - i a \sigma_- \right]}_{\hat{A}}$$

Recall: $H_{BSB} |g\rangle|0\rangle \longrightarrow |e\rangle|1\rangle$

$H_{BSB} |e\rangle|1\rangle \longrightarrow |g\rangle|0\rangle$ (Unitary)

Define this unitary: (on subspace $|g\rangle|0\rangle, |e\rangle|1\rangle$)

$U = e^{-iH_{BSB}t/\hbar}$. can use same formula:

$$U = \cos\left(\frac{\eta\sqrt{n}\Omega}{2}\right)\mathbb{I} - i\sin\left(\frac{\eta\sqrt{n}\Omega}{2}\right)\hat{A}$$

Show that

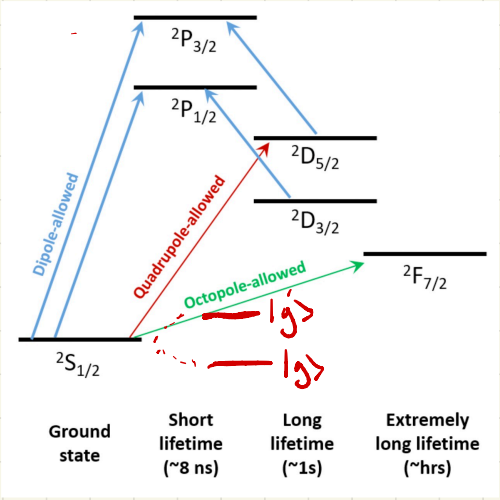
$U|e\rangle|0\rangle = |e\rangle|0\rangle$ (does nothing to 0-phonon excited state)

from form of $\hat{A}$; σ_+ annihilates, a annihilates $\rightarrow U|e\rangle|0\rangle = 0$

$\hat{A}|e\rangle|0\rangle = 0 \rightarrow$ stationary

Idea for phonon-controlled gate:

Need to use auxiliary level, $|g'\rangle$



Sequence:

① carrier $\pi/2$ $(|e\rangle, |g\rangle)$

② 2π - BSB $(|g\rangle|0\rangle \leftrightarrow |e\rangle|1\rangle)$

③ carrier $-\pi/2$ $(|e\rangle, |g\rangle)$

Exercise: Show the action of this gate on various states

Action:

$|g\rangle|1\rangle$

① $\longrightarrow \frac{1}{\sqrt{2}}(|g\rangle + |e\rangle)|1\rangle$

② $\longrightarrow \frac{1}{\sqrt{2}}(|g\rangle - |e\rangle)|1\rangle$

③ $\longrightarrow -|e\rangle|1\rangle$

$|e\rangle|1\rangle$

① $\longrightarrow \frac{1}{\sqrt{2}}(|e\rangle - |g\rangle)|1\rangle$

② $\longrightarrow -\frac{1}{\sqrt{2}}(|e\rangle + |g\rangle)|1\rangle$

③ $\longrightarrow -|g\rangle|1\rangle$

$|g\rangle|0\rangle$

① $\longrightarrow \frac{1}{\sqrt{2}}(|g\rangle + |e\rangle)|0\rangle$

② $\longrightarrow \frac{1}{\sqrt{2}}(|g\rangle + |e\rangle)|0\rangle$

③ → $|g\rangle|0\rangle$.

$|e\rangle|0\rangle$ ① → $\frac{1}{\sqrt{2}}(|e\rangle - |g\rangle)|0\rangle$

② → $\frac{1}{\sqrt{2}}(|e\rangle - |g\rangle)|0\rangle$

③ → $|e\rangle|0\rangle$

↓ phonon

$|0\,0\rangle$ etc. basis.

↑

elect.

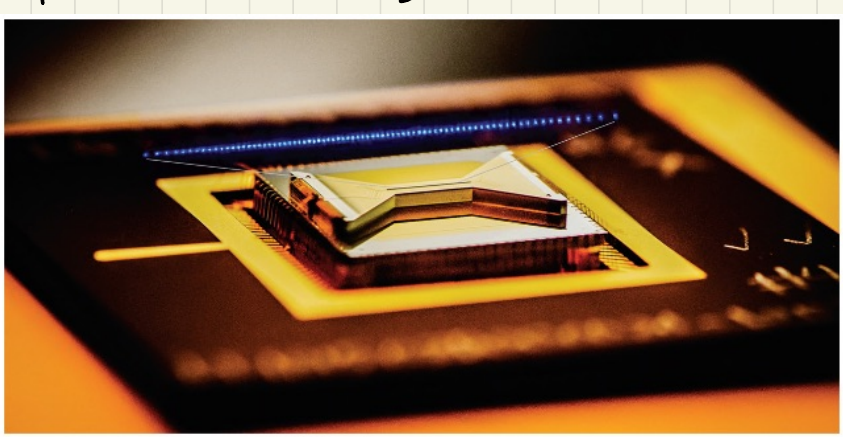
$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{pmatrix} = \text{CNOT.}$$ (modulo phase change on NOT gate)

✱ Note: States Comp. basis.

convention is: $|e\rangle \to |0\rangle$

$|g\rangle \to |1\rangle$

Multiple Ions (1D chain) More useful to have 2+ qubits!



Ions interact:

$$V = \underbrace{\sum_{m=1}^{N} \frac{M}{2} \nu^2 x_m^2}_{\text{harmonic trap}} + \underbrace{\sum_{\substack{m,n=1 \\ m \neq n}}^{N} \frac{Z^2 e^2}{8\pi\epsilon_0} \frac{1}{|x_m - x_n|}}_{\text{Coulomb interaction.}}$$

Z: deg. of ionisation.

M: mass of the ion

e: electron charge

ϵ_0: vacuum permitivity

ν: secular trap frequency

Assume we have small displacements

$$x_m = x_m^{(0)} + q_m$$

$x_m^{(0)}$: equilibrium positions; q_m: small displacement from zero.

$$\left.\frac{\partial V}{\partial x_m}\right|_{x_m = x_m^{(0)}} = 0$$

[Note: Additional detail in James et al. Appl. Phys. B, 66, 181 (1998)]

Results in Hamiltonian:

$$H = \frac{1}{2m}\sum_{m=1}^{N} p_m^2 + \frac{mv^2}{2}\sum_{m,n=1}^{N} A_{mn}\, q_m q_n$$

(p_m, q_m canonical momentum/position pair)

A_{mn} → $m \times n$ coupling matrix (real, symmetric)

non-negative → positive eigenvalues

Eigenvalue eqn: (global, normal mode coordinates)

$$\sum_{n=1}^{N} A_{mn}\, \beta_m^{(p)} = \mu_p\, \beta_n^{(p)} \quad \text{when } (p = 1, \ldots N)$$

Example: $N = 3$ (three ions)

$$\vec{\beta}^{(1)} = \frac{1}{\sqrt{3}}(1,1,1) \; ; \; \mu_1 = 1.$$

$$\vec{\beta}^{(2)} = \frac{1}{\sqrt{2}}(-1,0,1) \; ; \; \mu_2 = 3.$$

$$\beta^{(3)} = \frac{1}{\sqrt{6}}(1,-2,1) \; ; \; \mu_3 = 29/5$$

Collective Normal modes:

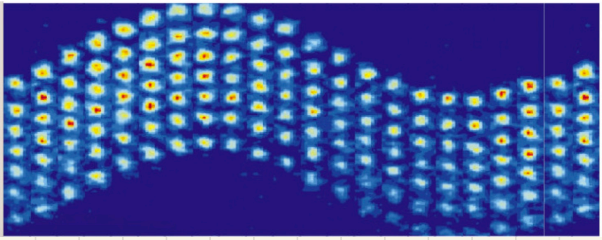
$$Q_p(t) = \sum_{n=1}^{N} \beta_n^{(p)} q_n(t) \quad \text{(in global coordinates)}$$

$$q_n(t) = \sum_{p=1}^{N} \beta_n^{(p)} Q_p(t) \quad \text{(in local coordinates)}.$$

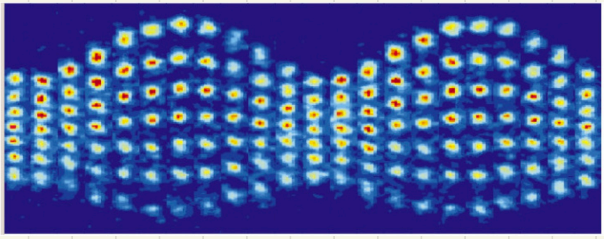
Characteristics of these modes:

- From form of β above, can see how this works out.

$p=1 \longrightarrow$ centre of mass (C.O.M.) mode



$p=2 \longrightarrow$ breathing mode



$p=3 \longrightarrow$ something funkier.

Frequency: $\nu_p = v\sqrt{\mu_p}$

Hamiltonian for the collective modes:

$$H = \frac{1}{2M}\sum_{p=1}^{N} P_p^2 + \frac{M}{2}\sum_{p=1}^{N} \nu_p^2 Q_p^2 \quad ; \quad P_p = M\dot{Q}_p$$

$$Q_p = \sqrt{\frac{\hbar}{2M\nu_p}}\left(b_p + b_p^\dagger\right)$$

$$P_p = \sqrt{\frac{\hbar M \nu_p}{2}}\left(b_p - b_p^\dagger\right)$$

Can use same approach as the single ion case to derive similar interaction Hamiltonian

↳ Result for C.O.M mode:

$$H_I^{(m)} = -i\hbar\eta\Omega_m \frac{}{2\sqrt{N}}\left(\sigma_+^{(m)} b_1 + \sigma_-^{(m)} b_1^\dagger\right)$$

- Interesting result!
- Can address a collective mode via interactions w/ single ion.
- Address ion separately w/ laser beams (coupled ions need not be adjacent)

How to use this for a CNOT between two ions?

Assume two ions share the same phonon mode (c.o.m. mode)

i.e. $|0\rangle_1|0\rangle_{ph}$; $|1\rangle_1|1\rangle_{ph}$; etc.

$|0\rangle_2|0\rangle_{ph}$ $|1\rangle_2|1\rangle_{ph}$; etc

So possible composite states: (start w/ $|0\rangle_{ph}$

$|0\rangle_1|0\rangle_2|0\rangle_{ph}$ (A) ; $|1\rangle_1|0\rangle_2|0\rangle$ (B) ; $|0\rangle_1|1\rangle_2|0\rangle$ (C) ; $|1\rangle_1|1\rangle_2|0\rangle$ (D)

Exercise: Show that the following sequence yields a CNOT gate:

Control is ion 1, target on ion 2.

1. BSB π-pulse on ion 1 (and shared phonon)
2. CNOT on ion 2 controlled by phonon
3. BSB π pulse on ion 1.

Showing this:

Recall BSB π-pulse: $(|1\rangle|0\rangle \rightarrow |0\rangle|1\rangle)$

$|g\rangle|0\rangle \rightarrow |e\rangle|1\rangle$

$|e\rangle|1\rangle \rightarrow -|g\rangle|0\rangle$

$|e\rangle|0\rangle \rightarrow |e\rangle|0\rangle$

Step 1:

(A) $|e\rangle|e\rangle|0\rangle \longrightarrow$ no effect.

(B) $|g\rangle|e\rangle|0\rangle \longrightarrow |e\rangle|e\rangle|1\rangle = B'$

(C) $|e\rangle|g\rangle|0\rangle \longrightarrow$ no effect.

(D) $|g\rangle|g\rangle|0\rangle \longrightarrow |e\rangle|g\rangle|1\rangle = D'$

Step 2:

$B' = |e\rangle|e\rangle|1\rangle \longrightarrow -|e\rangle|g\rangle|1\rangle$

$D' = |e\rangle|g\rangle|1\rangle \longrightarrow -|e\rangle|e\rangle|1\rangle$

Step 3:

$B' = -|e\rangle|g\rangle|1\rangle \longrightarrow |g\rangle|g\rangle|1\rangle = B''$

$D' = -|e\rangle|e\rangle|1\rangle \longrightarrow |g\rangle|e\rangle|1\rangle = D''$

In the end:

[qubit basis]

$|e\rangle|e\rangle \longrightarrow |e\rangle|e\rangle$; $[|0\rangle|0\rangle \longrightarrow |0\rangle|0\rangle]$

$|g\rangle|e\rangle \longrightarrow |g\rangle|g\rangle$; $[|1\rangle|0\rangle \longrightarrow |1\rangle|1\rangle]$

$|e\rangle|g\rangle \longrightarrow |e\rangle|g\rangle$ $[|0\rangle|1\rangle \longrightarrow |0\rangle|1\rangle]$

$|g\rangle|g\rangle \longrightarrow |g\rangle|e\rangle$ $[|1\rangle|1\rangle \longrightarrow |1\rangle|0\rangle]$

} Checks out

Outlook: (Hélène Perrin lecture notes)

Algorithms implemented with trapped ions

The recent implementations were done by Monroe's group (Maryland) or Blatt's group (Innsbruck).

- Spin and phase flip error detection up to 7 ions [Nigg et al. Science **345**, 302 (2014)]
- Deutsch-Jozsa / Bernstein-Vazirani algorithm: determines if a $N \to 1$ operation is either constant or balanced ($3 + 1$ ions + ancilla) [Debnath et al. Nature **536**, 63 (2016)]
- Quantum Fourier transform (QFT) (with 5 ions) [Debnath et al. ibid.]
- Grover's search algorithm (with 2 and 3 ions) [C. Figgatt et al., Nature Comm. **8**, 1918 (2017)]
- Shor algorithm (with 5 ions): $15 = 3 \times 5$ [Monz et al. Science **351**, 1068 (2016)]
- Variational quantum energy solver (VQE) applied to the ground state of H_2, LiH and H_2O (up to 11 ions) [C. Hempel et al., Phys. Rev. X **8**, 031022 (2018); Y. Nam et al., arXiv:1902.10171 (2019)]

DiVincenzo criteria with a trapped-ion quantum computer

- A scalable physical system with well characterised qubits. **Scalability?**
- The ability to initialize the state of the qubits to a simple fiducial state. **OK with laser cooling and optical pumping, < 1 ms**
- Long relevant decoherence times. **OK with microwave transitions or narrow-line optical transitions, > 100 ms**
- A universal set of quantum gates. **OK: single-qubit rotations with fidelity up to 99.9999% (microwave), CNOT up to 99.9%**
- A qubit-specific measurement capability. **OK with the shelving method, < 200 μs with fidelity 99.99%**

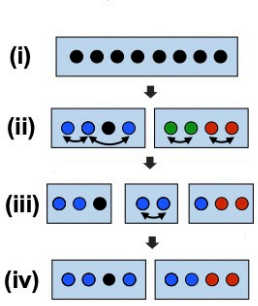
N.B. State preparation + readout with a fidelity 99.93% demonstrated, i.e. state preparation error $\sim 2 \times 10^{-4}$.

Source: Bruzewicz et al.



Efforts towards scalability

- Development of surface ion traps
- Having hundreds of addressable ions in a trap is 'easy'
- Difficult point: controlling their interation (with phonons)
- Possible solution: displace ions in a specific interaction zone
- Explored by Dave Wineland (NIST Boulder), Chris Monroe (Univ. Maryland and IonQ company), ...



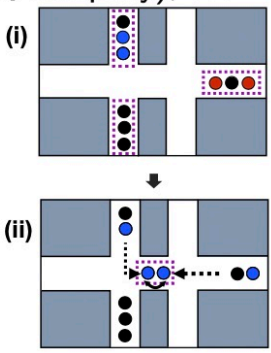


Fig. from Bruzewicz et al.

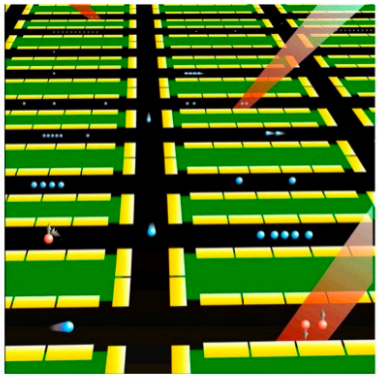


Fig. from NIST